

STA 445 S26 Assignment 4

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Load all the packages you will use in this chunk. Notice, I have set `include = FALSE` because I don't want to see you loading packages in the output.

Problem 1

For the following regular expression, explain in words what it matches on. Then add test strings to demonstrate that it in fact does match on the pattern you claim it does. Do at least 4 tests. Make sure that your test set of strings has several examples that match as well as several that do not. Make sure to remove the `eval=FALSE` so that the code and output shows in your pdf.

a. This regular expression matches: letter a

```
strings <- c("I have 2 apples", "My dog likes bones",  
             "My makeup is expensive", "The pool is empty")  
data.frame( string = strings ) %>%  
  mutate( result = str_detect(string, 'a') )
```

```
##           string result  
## 1      I have 2 apples  TRUE  
## 2    My dog likes bones FALSE  
## 3 My makeup is expensive  TRUE  
## 4    The pool is empty  FALSE
```

b. This regular expression matches: letter ab exactly

```
strings <- c("There is another dog barking", "I am absolutely happy",  
             "I'm able to run a mile", "No one is coming")  
data.frame( string = strings ) %>%  
  mutate( result = str_detect(string, 'ab') )
```

```
##           string result  
## 1 There is another dog barking FALSE  
## 2      I am absolutely happy  TRUE  
## 3    I'm able to run a mile  TRUE  
## 4      No one is coming  FALSE
```

- c. This regular expression matches: words with either letter a or b

```
strings <- c("Students made a rocket for class",
             "I produced the world of flowers", "They're not good students",
             "Your behavior is horrible")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '[ab]') )
```

```
##              string result
## 1 Students made a rocket for class    TRUE
## 2 I produced the world of flowers    FALSE
## 3      They're not good students    FALSE
## 4      Your behavior is horrible    TRUE
```

- d. This regular expression matches: letter a or b at the beginning of input string

```
strings <- c("apples are sweet", "blue is the color of the sky",
             "cross the intersection", "purple is my favorite color")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '^[ab]') )
```

```
##              string result
## 1      apples are sweet    TRUE
## 2 blue is the color of the sky    TRUE
## 3      cross the intersection    FALSE
## 4 purple is my favorite color    FALSE
```

- e. This regular expression matches: one or more digit, followed by any white space, followed by letter a or A

```
strings <- c("I always bring cinnamon buns", "I have 12ants",
             "There are 2 airplanes", "There are 3 Ambers in the class")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '\\d+\\s[aA]') )
```

```
##              string result
## 1 I always bring cinnamon buns    FALSE
## 2      I have 12ants    FALSE
## 3      There are 2 airplanes    TRUE
## 4 There are 3 Ambers in the class    TRUE
```

- f. This regular expression matches: one or more digit, followed by zero or more any white space, followed by letter a or A

```
strings <- c("1a is the classroom number", "3 amazing spots available",
             "I gave you 20 dollars afterclass", "I meant to say 1AA")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '\\d+\\s*[aA]') )
```

```
##              string result
## 1      1a is the classroom number    TRUE
## 2      3 amazing spots available    TRUE
## 3 I gave you 20 dollars afterclass FALSE
## 4      I meant to say 1AA    TRUE
```

g. This regular expression matches: any character zero or more times

```
strings <- c("Listening to music was great", "Only one person allowed.",
             "Keep the line going", "abc is 3 letters")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '.*') )
```

```
##              string result
## 1 Listening to music was great    TRUE
## 2   Only one person allowed.    TRUE
## 3   Keep the line going    TRUE
## 4   abc is 3 letters    TRUE
```

h. This regular expression matches: beginning of the input string has any 2 number or letter followed by bar

```
strings <- c("14 bars in the area", "we disembark on a journey",
             "unbar the ladder", "22bars")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '^\\w{2}bar') )
```

```
##              string result
## 1      14 bars in the area FALSE
## 2 we disembark on a journey FALSE
## 3      unbar the ladder    TRUE
## 4      22bars    TRUE
```

i. This regular expression matches: the literal string of foo.bar or beginning of the input string has any 2 number or letter followed by bar

```
strings <- c("foo.bar", "unbar the ladder", "we can foolbar", "foo is almost bar")
data.frame( string = strings ) %>%
  mutate( result = str_detect(string, '(foo\\.bar)|(^\\w{2}bar)') )
```

```
##              string result
## 1      foo.bar    TRUE
## 2 unbar the ladder    TRUE
## 3   we can foolbar FALSE
## 4 foo is almost bar FALSE
```

Problem 2

The following file names were used in a camera trap study. The S number represents the site, P is the plot within a site, C is the camera number within the plot, the first string of numbers is the YearMonthDay and the second string of numbers is the HourMinuteSecond.

```
file.names <- c( 'S123.P2.C10_20120621_213422.jpg',  
                'S10.P1.C1_20120622_050148.jpg',  
                'S187.P2.C2_20120702_023501.jpg')
```

Produce a data frame with columns corresponding to the **site**, **plot**, **camera**, **year**, **month**, **day**, **hour**, **minute**, and **second** for these three file names. So we want to produce code that will create the data frame:

Site	Plot	Camera	Year	Month	Day	Hour	Minute	Second
S123	P2	C10	2012	06	21	21	34	22
S10	P1	C1	2012	06	22	05	01	48
S187	P2	C2	2012	07	02	02	35	01

```
split.names <- data.frame(str_split_fixed(file.names, pattern="[_.]", n=Inf)) %>%  
  mutate(Site = X1,  
         Plot = X2,  
         Camera = X3,  
         Year = str_sub(X4, start=1, end = 4),  
         Month = str_sub(X4, start=5, end=6),  
         Day = str_sub(X4, start=7, end= 8),  
         Hour = str_sub(X5, start=1, end =2),  
         Minute = str_sub(X5, start=3, end = 4),  
         Second = str_sub(X5, start=5, end=6))%>%  
  select(Site, Plot, Camera, Year, Month, Day, Hour, Minute, Second)  
  
head(split.names)
```

```
##   Site Plot Camera Year Month Day Hour Minute Second  
## 1 S123  P2     C10 2012   06  21   21     34     22  
## 2  S10   P1      C1 2012   06  22   05     01     48  
## 3 S187  P2     C2 2012   07  02   02     35     01
```

3. The full text from Lincoln's Gettysburg Address is given below. Calculate the mean word length
Note: consider 'battle-field' as one word with 11 letters). Goal: 271 words, mean of words = 4.239852

```
Gettysburg <- 'Four score and seven years ago our fathers brought forth on this continent, a new  
  
gettysburg <- str_replace_all(Gettysburg, pattern='[.,]', replacement = ' ') %>%  
  str_replace_all(pattern='[-]', replacement = '') %>%  
  str_trim(side="right") %>%  
  str_split(pattern='\\s+')  
  
gettysburg
```

##	[[1]]					
##	[1]	"Four"	"score"	"and"	"seven"	"years"
##	[6]	"ago"	"our"	"fathers"	"brought"	"forth"
##	[11]	"on"	"this"	"continent"	"a"	"new"
##	[16]	"nation"	"conceived"	"in"	"Liberty"	"and"
##	[21]	"dedicated"	"to"	"the"	"proposition"	"that"
##	[26]	"all"	"men"	"are"	"created"	"equal"
##	[31]	"Now"	"we"	"are"	"engaged"	"in"
##	[36]	"a"	"great"	"civil"	"war"	"testing"
##	[41]	"whether"	"that"	"nation"	"or"	"any"
##	[46]	"nation"	"so"	"conceived"	"and"	"so"
##	[51]	"dedicated"	"can"	"long"	"endure"	"We"
##	[56]	"are"	"met"	"on"	"a"	"great"
##	[61]	"battlefield"	"of"	"that"	"war"	"We"
##	[66]	"have"	"come"	"to"	"dedicate"	"a"
##	[71]	"portion"	"of"	"that"	"field"	"as"
##	[76]	"a"	"final"	"resting"	"place"	"for"
##	[81]	"those"	"who"	"here"	"gave"	"their"
##	[86]	"lives"	"that"	"that"	"nation"	"might"
##	[91]	"live"	"It"	"is"	"altogether"	"fitting"
##	[96]	"and"	"proper"	"that"	"we"	"should"
##	[101]	"do"	"this"	"But"	"in"	"a"
##	[106]	"larger"	"sense"	"we"	"can"	"not"
##	[111]	"dedicate"	"we"	"can"	"not"	"consecrate"
##	[116]	"we"	"can"	"not"	"hallow"	"this"
##	[121]	"ground"	"The"	"brave"	"men"	"living"
##	[126]	"and"	"dead"	"who"	"struggled"	"here"
##	[131]	"have"	"consecrated"	"it"	"far"	"above"
##	[136]	"our"	"poor"	"power"	"to"	"add"
##	[141]	"or"	"detract"	"The"	"world"	"will"
##	[146]	"little"	"note"	"nor"	"long"	"remember"
##	[151]	"what"	"we"	"say"	"here"	"but"
##	[156]	"it"	"can"	"never"	"forget"	"what"
##	[161]	"they"	"did"	"here"	"It"	"is"
##	[166]	"for"	"us"	"the"	"living"	"rather"
##	[171]	"to"	"be"	"dedicated"	"here"	"to"
##	[176]	"the"	"unfinished"	"work"	"which"	"they"
##	[181]	"who"	"fought"	"here"	"have"	"thus"
##	[186]	"far"	"so"	"nobly"	"advanced"	"It"
##	[191]	"is"	"rather"	"for"	"us"	"to"
##	[196]	"be"	"here"	"dedicated"	"to"	"the"
##	[201]	"great"	"task"	"remaining"	"before"	"us"
##	[206]	"that"	"from"	"these"	"honored"	"dead"
##	[211]	"we"	"take"	"increased"	"devotion"	"to"
##	[216]	"that"	"cause"	"for"	"which"	"they"
##	[221]	"gave"	"the"	"last"	"full"	"measure"
##	[226]	"of"	"devotion"	"that"	"we"	"here"
##	[231]	"highly"	"resolve"	"that"	"these"	"dead"
##	[236]	"shall"	"not"	"have"	"died"	"in"
##	[241]	"vain"	"that"	"this"	"nation"	"under"

```
## [246] "God"      "shall"    "have"     "a"        "new"
## [251] "birth"    "of"       "freedom"  "and"      "that"
## [256] "government" "of"      "the"      "people"   "by"
## [261] "the"      "people"   "for"      "the"      "people"
## [266] "shall"    "not"      "perish"   "from"     "the"
## [271] "earth"
```

```
str_length(gettysburg[[1]]) %>%
  mean()
```

```
## [1] 4.239852
```